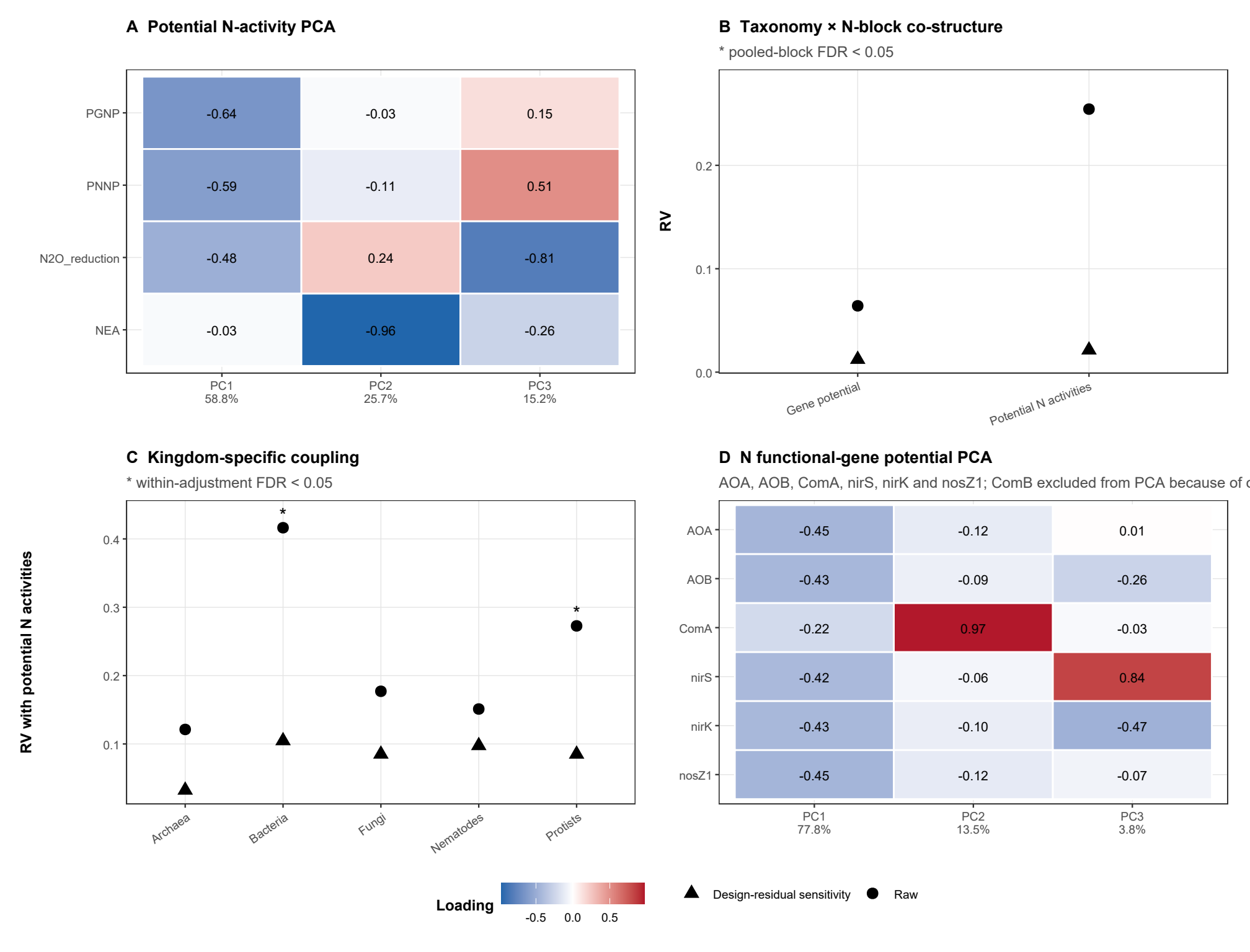
A Potential N-activity PCA
PGNP
PNNP
N2O_reduction
NEA
-0.64
-0.03
0.15
-0.59
-0.11
0.51
-0.48
0.24
-0.81
-0.03
-0.96
-0.26
PC1
58.8%
PC2
25.7%
PC3
15.2%
B Taxonomy × N-block co-structure
* pooled-block FDR < 0.05
RV
0.0
0.1
0.2
Gene potential
Potential N activities
C Kingdom-specific coupling
* within-adjustment FDR < 0.05
RV with potential N activities
0.1
0.2
0.3
0.4
*
*
Archaea
Bacteria
Fungi
Nematodes
Protists
D N functional-gene potential PCA
AOA, AOB, ComA, nirS, nirK and nosZ1; ComB excluded from PCA because of
AOA
AOB
ComA
nirS
nirK
nosZ1
-0.45
-0.12
0.01
-0.43
-0.09
-0.26
-0.22
0.97
-0.03
-0.42
-0.06
0.84
-0.43
-0.10
-0.47
-0.45
-0.12
-0.07
PC1
77.8%
PC2
13.5%
PC3
3.8%
Loading
-0.5
0.0
0.5
Design-residual sensitivity
Raw